

# Prime Numbers and the Fundamental Theorem of Arithmetic

- **Definition 1**

An integer  $p > 1$  is a **prime number** if its only positive divisors are 1 and  $p$ . An integer  $n > 1$  that is not prime is called **composite**.

- **Example 1.1**

The first prime numbers are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, \dots$$

The numbers

$$4, 6, 8, 9, 10, 12, \dots$$

are composite.

Note that 1 is neither prime nor composite.

- **Proposition 1.2**

An integer  $n > 1$  is composite, if and only if,  $n = ab$  for some  $a, b \in \mathbb{N}$  with  $1 < a < n$  and  $1 < b < n$ .

- **Proof 1.2.1**

( $\Rightarrow$ ) Suppose that  $n$  is composite, then it has a positive divisor  $a$  with  $a \neq 1, n$ . Hence there exists  $b \in \mathbb{N}$  such that  $n = ab$  with  $1 < a < n$  and  $1 < b < n$ , because if  $b = 1$  then  $a = n$ .

( $\Leftarrow$ ) Suppose that  $n = ab$  for some  $a, b \in \mathbb{N}$  with  $1 < a < n$  and  $1 < b < n$ . Then such a factorization makes  $a$  a divisor of  $n$  with  $a \neq 1, n$ . Therefore  $n$  is composite.

- **Lemma 1.3**

Let  $p$  be a prime and let  $a \in \mathbb{Z}$ . If  $p \nmid a$ , then there exist  $x, y \in \mathbb{Z}$  such that  $ax + py = 1$ .

- **Proof 1.3.1**

Consider the set

$$S = \{ax + py : x, y \in \mathbb{Z} \wedge ax + py > 0\}.$$

Since  $p \nmid a$ , we have that  $a \neq 0$ . Therefore  $|a| \in S$  because  $|a| = a \cdot 1 + p \cdot 0 > 0$  if  $a > 0$ , or  $|a| = a \cdot (-1) + p \cdot 0 > 0$  if  $a < 0$ .

Now, as  $S \subseteq \mathbb{N}$  and  $S \neq \emptyset$ , by the well-ordering axiom  $S$  has a first element  $d$ . Therefore  $d = ax_0 + py_0 > 0$ . If we divide  $a$  by  $d$ , then by the division algorithm we have that

$$a = dq + r \text{ with } 0 \leq r < |d| = d.$$

Hence

$$r = a - dq = a - (ax_0 + py_0)q = a(1 - x_0q) + p(-y_0q).$$

If  $r > 0$ , then  $r \in S$ , which contradicts the minimality of  $d$ . Therefore we obtain that  $r = 0$  and  $a = dq$ , that is,  $d \mid a$ . Using the same argument, dividing  $p$  by  $d$  shows that  $d \mid p$ . Now, as  $p$  is prime we have that  $d = 1 \vee d = p$ , but as  $p \nmid a$  we have that  $d = 1$ . Therefore we obtain that  $ax_0 + py_0 = 1$ .

- **Corollary 1.4**

(1) Let  $p$  be a prime and let  $a, b \in \mathbb{Z}$ . If  $p \mid ab$ , then  $p \mid a \vee p \mid b$ .

- **Proof 1.4.1**

Suppose that  $p \nmid a$ , then by Lemma 1.3 there exist  $x, y \in \mathbb{Z}$  such that  $ax + py = 1$ , hence

$$abx + pby = b.$$

Since  $p \mid ab$ , there exists  $k \in \mathbb{Z}$  such that  $ab = kp$ , so

$$b = abx + pby = kpx + pby = (kx + by)p \text{ with } kx + by \in \mathbb{Z}.$$

Therefore  $p \mid b$ .

(2) Let  $p$  be a prime and  $a_1, a_2, \dots, a_n \in \mathbb{Z}$ . If  $p \mid (a_1 a_2 \cdots a_n)$ , then  $p \mid a_i$  for some  $1 \leq i \leq n$ . In particular, if all the  $a_j$  are prime, then  $p = a_i$  for some  $1 \leq i \leq n$ .

■ **Proof 1.4.2**

By induction.

1. Base step

If  $n = 1$  then the result is true.

2. Inductive step

Suppose that the result is true for  $n$ .

$$\begin{aligned} p \mid (a_1 a_2 \cdots a_n a_{n+1}) &\implies p \mid (a_1 a_2 \cdots a_n) a_{n+1} \\ &\implies p \mid (a_1 a_2 \cdots a_n) \vee p \mid a_{n+1} \\ &\implies p \mid a_i \text{ for some } 1 \leq i \leq n \vee p \mid a_{n+1} \\ &\implies p \mid a_j \text{ for some } 1 \leq j \leq n+1. \end{aligned}$$

Therefore the result is true for  $n+1$ .

If all the  $a_j$  are prime and  $p \mid a_i$ , then by definition of a prime we have that  $p = a_i$ .

- **Theorem 2 (Fundamental theorem of arithmetic)** Every integer  $n \geq 2$  can be written in a unique form up to rearrangements as a product

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$$

where  $p_1, p_2, \dots, p_r$  are distinct primes and  $\alpha_1, \alpha_2, \dots, \alpha_r \in \mathbb{N}$ . For each  $i$ , the notation  $\mathcal{U}_{p_i}(n) = \alpha_i$  denotes the exponent of the prime  $p_i$  in the factorization of  $n$ .

○ **Proof 2.1**

1. Existence

Consider the set

$$A = \{m \in \mathbb{N} : m \geq 2 \wedge m \text{ cannot be written as a product of primes}\}.$$

If  $A \neq \emptyset$ , then by the well-ordering axiom  $A$  has a first element  $n$ . Then  $n$  is not a prime. As  $n$  is composite, there exist  $a, b \in \mathbb{N}$  such that  $n = ab$  with  $1 < a < n$  and  $1 < b < n$ . As  $n$  is the first element of  $A$ , we have that  $a$  and  $b$  can be written as a product of primes, and therefore  $n$  can be written as a product of primes, which is a contradiction because  $n \in A$ . Therefore  $A = \emptyset$  and thus every integer  $\geq 2$  can be written as a product of prime numbers.

2. Uniqueness

Suppose that the result is false, that is, there exists an integer  $\geq 2$  that can be written in two different forms as a product of primes. Then the set

$$B = \{m \in \mathbb{N} : m \geq 2 \wedge m \text{ can be written in two different forms as a product of primes}\}$$

is nonempty, then by the well-ordering axiom  $B$  has a first element  $n$ , then

$$p_1 p_2 \cdots p_r = n = q_1 q_2 \cdots q_s.$$

Since  $p_1 \mid q_1 q_2 \cdots q_s$ , then  $p_1 \mid q_i$  for some  $1 \leq i \leq s$  by Corollary 1.4(2), and by commutativity of the product we may assume that  $i = 1$  and so  $p_1 = q_1$ . Hence by the cancellation law we have

$$p_2 p_3 \cdots p_r = q_2 q_3 \cdots q_s < n.$$

Therefore  $p_2 p_3 \cdots p_r$  and  $q_2 q_3 \cdots q_s$  are two different factorizations of a number  $< n$ , which contradicts the minimality of  $n$ . Therefore  $B = \emptyset$ , that is, every integer  $\geq 2$  can be written in a unique form as a product of primes.

○ **Example 2.2**

Find the prime factorization of 224 and 1260.

$$224 = 2^5 \cdot 7 \quad 1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7.$$

◦ **Corollary 2.3**

Every integer  $n \leq -2$  can be written in a unique form up to rearrangements as a product

$$n = (-1)p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}$$

where  $p_1, p_2, \dots, p_r$  are distinct primes and  $\alpha_1, \alpha_2, \dots, \alpha_r \in \mathbb{N}$ .

▪ **Proof 2.3.1**

Since  $n \leq -2$  then  $-n \geq 2$ , and by the fundamental theorem of arithmetic

$$-n = p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}$$

where  $p_1, p_2, \dots, p_r$  are distinct primes and  $\alpha_1, \alpha_2, \dots, \alpha_r \in \mathbb{N}$ .

Hence

$$n = (-1)p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}.$$

◦ **Proposition 2.4**

Suppose  $x, y \in \mathbb{N}$ , then  $\mathcal{U}_p(xy) = \mathcal{U}_p(x) + \mathcal{U}_p(y)$  for any prime  $p$ .

▪ **Proof 2.4.1**

Consider the prime factorizations of  $x$  and  $y$ . Let

$$\begin{aligned} x &= p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r} & \alpha_k &\in \mathbb{N}_0, \\ y &= p_1^{\beta_1}p_2^{\beta_2}\cdots p_r^{\beta_r} & \beta_k &\in \mathbb{N}_0. \end{aligned}$$

Then we have

$$xy = p_1^{\alpha_1+\beta_1}p_2^{\alpha_2+\beta_2}\cdots p_r^{\alpha_r+\beta_r}.$$

By the uniqueness of the prime factorization of  $xy$ , for each  $p = p_k$ , we have that

$$\mathcal{U}_p(xy) = \alpha_k + \beta_k = \mathcal{U}_p(x) + \mathcal{U}_p(y).$$

• **Theorem 3 (Euclid's theorem)**

The set of prime numbers is infinite.

◦ **Proof 3.1**

Suppose that there are only finitely many primes  $p_1, p_2, \dots, p_n$  and consider

$$N = p_1p_2\cdots p_n + 1.$$

Since  $N > 1$ , by the fundamental theorem of arithmetic we have a prime divisor  $q$  of  $N$ . Then  $q = p_i$  for some  $1 \leq i \leq n$ , hence  $q$  divides both  $N$  and  $p_1p_2\cdots p_n$ , and therefore  $q$  divides  $N - p_1p_2\cdots p_n = 1$ , which is impossible because  $q$  is a prime. Therefore there are infinitely many primes.